Experiment - 8

In [1]:
```
import cv2
import numpy as np

# Step 1: Open video
cap = cv2.VideoCapture("videoo.mp4")

# Step 2: Read first frame
ret, old_frame = cap.read()
old_gray = cv2.cvtColor(old_frame, cv2.COLOR_BGR2GRAY)

# Step 3: Detect feature points using Shi-Tomasi
p0 = cv2.goodFeaturesToTrack(old_gray, 100, 0.3, 7)

while True:

    # Step 4: Read next frame
    ret, frame = cap.read()
    if not ret:
        break

    gray = cv2.cvtColor(frame, cv2.COLOR_BGR2GRAY)

    # Step 5: Apply Lucas-Kanade Optical Flow
    p1, status, _ = cv2.calcOpticalFlowPyrLK(
        old_gray, gray, p0, None
    )

    # Step 6: Draw motion trajectories
    for new, old in zip(p1[status == 1], p0[status == 1]):
        x1, y1 = new.ravel()
        x2, y2 = old.ravel()

        cv2.arrowedLine(
            frame,
            (int(x2), int(y2)),
            (int(x1), int(y1)),
            (0, 255, 0), 2
        )

        # Calculate motion
        dx = x1 - x2
        dy = y1 - y2
        speed = np.sqrt(dx**2 + dy**2)

    # Step 7: Display tracking result
    cv2.putText(frame, "Object Tracking",
                (20, 40), cv2.FONT_HERSHEY_SIMPLEX,
                1, (0, 255, 255), 2)

    cv2.imshow("Optical Flow Tracking", frame)

    # Step 8: Update previous frame and points
    old_gray = gray
    p0 = p1[status == 1].reshape(-1, 1, 2)

    # Step 9: Press Q to exit
```

```
    if cv2.waitKey(30) & 0xFF == ord('q'):
        break

# Step 10: Release resources
cap.release()
cv2.destroyAllWindows()
```

Questions and Answers

1. How does object tracking differ from object detection?

Object detection identifies objects in a single image or video frame and determines their location, usually using bounding boxes. Object tracking follows the identified object across multiple consecutive frames. Therefore, detection finds **what and where an object is**, while tracking determines **how the object moves over time**.

2. Explain how Optical Flow can be used for real-time object tracking.

Optical Flow calculates the apparent movement of pixels or feature points between consecutive video frames. Important feature points can be selected and tracked using the **Lucas-Kanade algorithm**. By continuously updating their positions, the movement, direction, and trajectory of an object can be estimated in real time.

3. What is the role of Shi-Tomasi Corner Detection in the Lucas-Kanade Optical Flow algorithm?

Shi-Tomasi Corner Detection is used to find **strong and distinctive feature points** that can be tracked reliably. These points are given as input to the Lucas-Kanade Optical Flow algorithm. Lucas-Kanade then estimates the movement of these points between consecutive frames, making object tracking more efficient.

4. Why is Optical Flow suitable for motion analysis in videos?

Optical Flow provides information about the **direction and magnitude of motion** between consecutive frames. It can detect how objects, surfaces, or the camera are moving without processing every frame independently for object detection. This makes it useful for tracking objects and analyzing movement patterns in video sequences.

5. What challenges arise while tracking fast-moving or partially occluded objects?

Fast-moving objects may move a large distance between two frames, making it difficult to find their corresponding feature points. Partial occlusion can hide important features and cause the tracker to lose the object. **Motion blur, lighting changes, background similarity, and camera movement** can also reduce tracking accuracy.

6. Compare Optical Flow-based tracking with deep learning-based object tracking methods.

Optical Flow-based tracking estimates movement using pixel or feature changes and usually requires less computational power. Deep learning-based tracking uses trained neural networks to identify and track objects, providing better performance in complex

scenes. However, deep learning methods generally require **more computing resources, training data, and model preparation**.

7. How can Optical Flow be used in traffic monitoring and autonomous driving systems?

Optical Flow can estimate the movement of **vehicles, pedestrians, and other objects** from consecutive video frames. In traffic monitoring, it can help determine vehicle direction, movement, and approximate speed. In autonomous driving, it can support obstacle detection, motion estimation, and understanding the movement of surrounding objects.

8. Explain the effect of camera motion on Optical Flow estimation.

When a camera moves, stationary objects also appear to move in the video. As a result, Optical Flow may detect motion throughout the scene even though some objects are actually stationary. This makes it difficult to separate **camera motion from object motion**, so techniques such as camera motion compensation may be required.

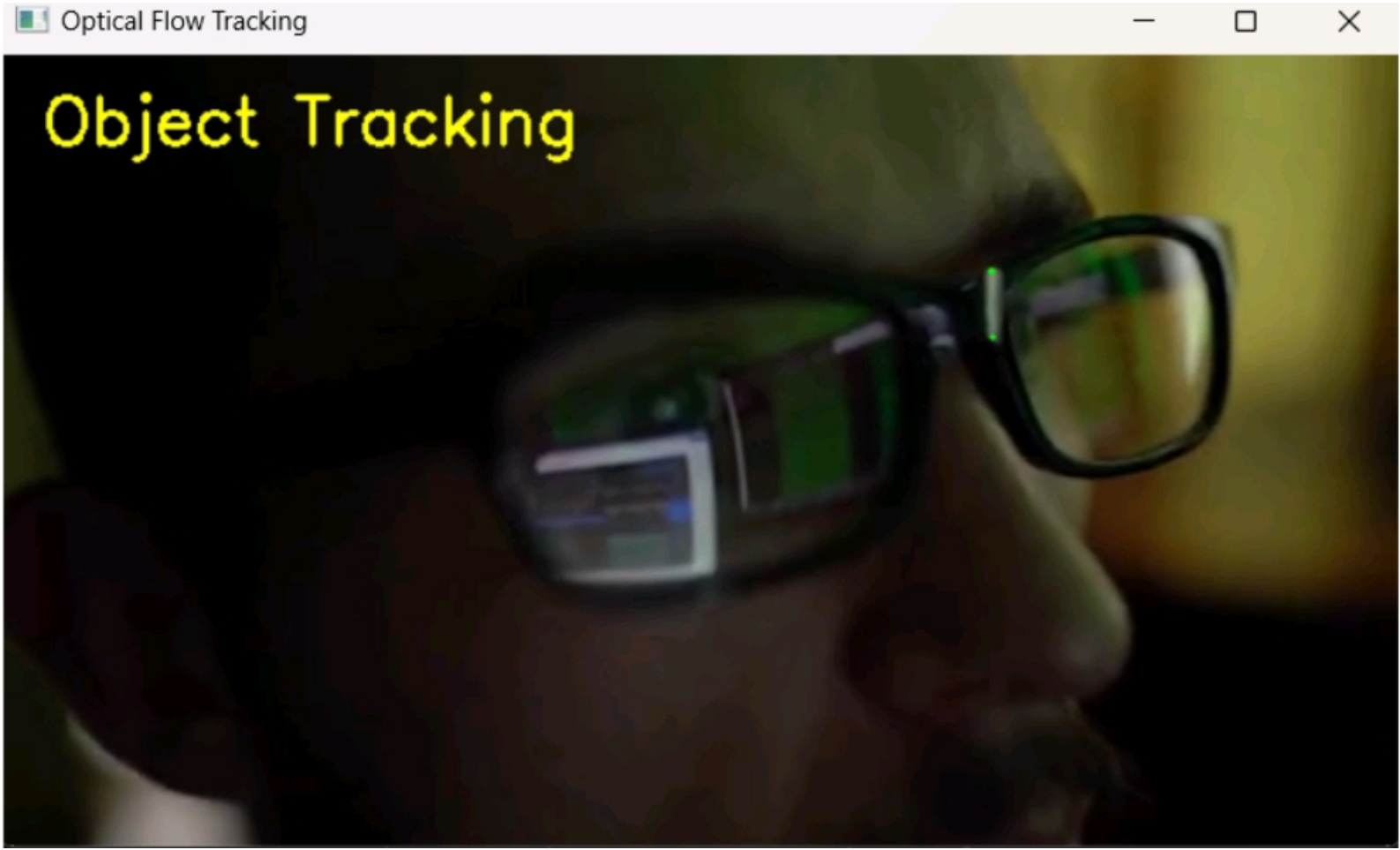
9. Mention five real-world applications where motion analysis using Optical Flow is commonly employed.

Five important applications of Optical Flow are:

1. **Traffic monitoring** – Tracking vehicles and analyzing traffic movement.
2. **Video surveillance** – Detecting and tracking moving objects.
3. **Robotics** – Supporting navigation and obstacle avoidance.
4. **Sports analytics** – Tracking players and ball movement.
5. **Human activity recognition** – Analyzing gestures and body movements.

10. How can Optical Flow improve the performance of surveillance, robotics, and human activity recognition systems?

Optical Flow provides information about **movement, direction, and speed**, which helps systems understand dynamic environments. In surveillance, it can detect unusual movements and track objects. In robotics, it supports navigation and obstacle avoidance. In human activity recognition, it helps identify actions by analyzing patterns of body movement.


Optical Flow Tracking
Object Tracking